USB-C PD -> fixed 9 V Arduino_NMR_5.4 power input

Standalone GPIO mode: two resistors set the request. No I2C, no MCU, no firmware, no switching regulator.

D20
SMAJ24A
place D20 physically at J10
J10
USB4105-GF-A-120
VBUS
CC1
CC2
D+/D-, SBU unrouted:
BC1.2 detection only
SHIELD
GND
SH
R27
1M
C27
4.7nF
shield: DC-isolated, RF short
R20
10R
C20
1uF/50V
C21
1uF/50V
U20
HUSB238A-BB001-QN16R
VDD is a supply INPUT, left unpowered:
VBUS powers the core. The FAULT push-pull
level references VDD - see README, test 9.
Stuffing it selects I2C mode.
ADDR/ORIENT must FLOAT - no footprint, or DNP.
EN_N + FLGIN to GND = enabled
U20_GATE
R24
2.26k 1%
R23
6.04k 1%
-> requests 9 V
an OPEN R23 requests 28 V
requests 1.5 A
R25
1M
R26
1M
FAULT/OUT2 goes HIGH when no source PDO covers the request, or on
OVP / UVP / OTP. R28 is Q21's base resistor. If the LED never
lights, see README test 9 - VDD is unpowered and sets VOH here.

Pass switch and bulk

U20_GATE
Q20_GATE
R21
5.1k
R22
100k
Q20
DMP3013SFV-7
+9V_SYS
VBUS
ALL bulk sits behind Q20: Type-C caps sink
capacitance near 10 uF before negotiation.
All three source pins and all five drain pins are tied -
the whole 800 mA goes through them.
C22
22uF/25V
C23
22uF/25V
C24
100nF/50V
D21
SMAJ11A

IC25
BD33GA5WEFJ-E2
C12
10 nF
C25
10 nF
+3V3
CTL_3V3 (PI13)
IC7
BD80GC0VEFJ-ME2
C26
10 nF
C28
10 nF
+8V
CTL_8V (PI14)

Overvoltage interlock - trips at 10.0 V, survives 28 V

Independent of U20. BB001's factory SNK_PDO2 is 28 V / 3 A, and U20's own OVP is referenced to the voltage it requested, so it will not fire.

VBUS
R30
30.1k 1%
R31
10.0k 1%
R35
1k
R34
6.8k
D22
BZX84C10
Q22
BSS84
Q20_GATE
U21
TL431A
C30
10nF/50V
D22 is not optional: without it Q22 sees VGS = -25.5 V at a
28 V fault. It clamps VGS to -10 V and parks U21's cathode
at VBUS - 10 V, which is above the 7.0 V REF tap.
Vtrip = 2.495 x (R30 + R31) / R31 = 10.00 V
worst case 9.76 - 10.25 V. The 9 V PDO high corner is
9.45 V, so 310 mV of margin.
Sense VBUS, not the load: opening Q20 does not remove
the trip condition, so no hysteresis is needed.

+3V3
390
R37
LED3
RED
+8V
2.2k
R36
RED
LED2
+9V_SYS
R29
2.2k
LED1
RED

J4
Conn_02x05_Odd_Even
J2
Conn_01x12
TTL1
TTL2
TTL3
TTL4
TTL5
CTL_8V (PI14)
CTL_3V3 (PI13)
+9V_SYS

BOM, footprints and test matrix in ../BOM.md and ../README.md
Lower block: hardware overvoltage interlock, trips at 10.0 V.
HUSB238A-BB001, GPIO mode. No I2C, no MCU, no firmware.
Arduino_NMR_5.4 - NMR spectrometer

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